

Résumé of KRITHAGHO CHAKMA

📍 Masjidul Ekbar Eidgah Maidan, Mirpur-1, Dhaka-1216, Bangladesh
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CAREER OBJECTIVE

I possess a proficient understanding of DC/AC circuit analysis and design, digital electronics, solid state device physics, C/C++, and Cadence Virtuoso Tool. I am seeking hands-on experience in implementing device physics and knowledge within the relevant industry, such as electronics, power and telecommunication, to enhance my technical proficiency. My aspiration is to advance to acquire lots of expertise in my field, contributing as a valuable expert in the technology sector in Bangladesh.

ACADEMIC QUALIFICATION

Bachelor of Science | *Major: Electronics, Minor: Power* Jan. 13, 2019 – April 04, 2024
CGPA: 3.23 out of 4.00 University of Dhaka
Electrical and Electronic Engineering Dhaka - 1000, Bangladesh
• **Key Undergraduate Courses:** VLSI I & Laboratory, Semiconductor Technology, Microprocessor and Interfacing, Digital Electronics, Analog & Digital Circuit Simulation, Electronic Devices.

RESEARCH INTERESTS

- Quantum Information Hardware
- Advanced Solar Cells
- Opto-Electronics

STANDARDIZED TESTS

- **GRE:** 332 (Quant: 165, Verbal: 165, AWA: 2)
- **IELTS:** 7 (R: 6.5, S: 7, L: 6.5, W: 7)

PROFESSIONAL EXPERIENCE

Subject Program Senior Executive, ICT (*Promoted from Executive*) July, 2024 – Present
Shikho Technologies Bangladesh Limited Dhaka, Bangladesh
• Authored '*Shikho FUTUREBook*', a comprehensive ICT textbook, synthesizing complex computing concepts into accessible academic materials.
• Managed large-scale educational databases and structured ICT curricula, ensuring high technical accuracy and efficient data retrieval.

PUBLICATIONS

- J. A. Joya, M. A. Hossain, **K. Chakma**, and S. Hussain, "Numerical study of chalcopyrite **n – CuAlSe₂/p – CuGaSe₂** solar cell with an additional **CuTiTe₂** back absorber layer using a SCAPS-1D simulator and machine learning approach," *Semiconductor Science and Technology*, vol. 41, p. 025006, Feb. 2026.
[DOI: 10.1088/1361-6641/ae3f4b](https://doi.org/10.1088/1361-6641/ae3f4b)
Determined optimal cell parameters (efficiency, open circuit voltage, current density, fill factor) through extensive parametric variation and proposed a novel absorber layer configuration, achieving a remarkable power conversion efficiency of 30.6%.

RESEARCH EXPERIENCE & ACADEMIC PROJECTS

- **Numerical study of copper chalcopyrite semiconductor-based hetero-junction solar cell.**
[Advisor: Dr. Sakhawat Hussain] [[Undergraduate Thesis](#)] [[Link](#)] [2024]
Utilized SCAPS-1D simulation software to optimize solar cell parameters by varying layer thickness, doping concentration, and defects. Investigated the effect of adding a p–CuTiTe₂ layer to the n–CuAlSe₂/p–CuGaSe₂ structure to drastically enhance performance.

- **Design and verification of an 8-bit Up/Down counter.** [\[Academic Project\]](#) [\[Link\]](#) [\[2023\]](#)
Executed full digital design flow using Verilog HDL and Cadence Virtuoso, including testbench creation, logic synthesis, and full physical design/layout of the counter system.

TECHNICAL SKILLS

Programming: Verilog, C/C++, Python, Assembly Language

Scientific: COMSOL Multiphysics, Cadence Virtuoso

Analytical: SCAPS-1D, Origin Pro, MATLAB

Document Creation & Resource Management: Microsoft Office 365 Suite, Google Workspace Applications, $\text{\LaTeX}$

AWARDS AND ACHIEVEMENTS

- Securing 3rd Position in the science olympiad of **38th National Science and Technology Fair - 2017** organized by the *District Administration of Khagrachari Hill District* (2017)
- Government scholarship under *General* category based on the result of **Secondary School Certificate Examination - 2016** from the *Board of Intermediate and Secondary Education (Chittagong)*, *Ministry of Education, Bangladesh* (2016)
- Honorary award for winning **Srijonshil Medha Onneshon Protijogita - 2015** in the subject of *Mathematics and Computer* in 9th - 10th group at the district level (2015)

EXTRACURRICULAR ACTIVITIES

Secretary	2022 – 2023
IEEE Electron Devices Society Student Branch Chapter University of Dhaka	Dhaka - 1000, Bangladesh
Student Representative	2023
IEEE Photonics Society Bangladesh Chapter	Dhaka, Bangladesh
Program Coordinator	2021 – 2022
IEEE Electron Devices Society Student Branch Chapter University of Dhaka	Dhaka - 1000, Bangladesh

REFERENCES

Dr. Sakhawat Hussain Associate Professor Department of Electrical and Electronic Engineering University of Dhaka ☎ +880 171686552 ✉ sakhawat@du.ac.bd	Dr. Mainul Hossain Assistant Professor Department of Electrical and Electronic Engineering University of Dhaka ☎ +880 1782383782 ✉ mainul.eee@du.ac.bd
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